

Sampling, Guidance, and Observation Design in FM4PDE: A Systematic Ablation Study on Three PDE Problems

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Abstract

We study how guidance composition, loss-evaluation state, sampler phase, time discretization, observation design, and residual construction affect physics-guided flow-matching inference on Poisson, Burgers, and non-bounded Navier–Stokes problems. After resolving historical reruns, the analysis contains 274 unique successful configurations: 93 for Poisson, 85 for Burgers, and 96 for Navier–Stokes. Observation guidance is the dominant source of reconstruction accuracy on all three problems, whereas PDE-only guidance performs similarly to no guidance. Purely deterministic sampling is numerically unstable, while a stochastic-to-deterministic schedule can reduce the reconstruction score by 24%, 66%, and 15% on Poisson, Burgers, and Navier–Stokes, respectively. These accuracy gains are not uniformly physical: the best Poisson and Navier–Stokes schedules increase the reported PDE residual by factors of 166 and 2.4. Observation sparsity also has strongly problem-dependent effects, with a sharp failure below 500 observations for Poisson and below approximately 250–500 observations for Navier–Stokes, while Burgers degrades more smoothly. A tighter gradient clip prevents the Navier–Stokes Hermite residual from producing NaN/Inf, but 88 of 100 steps remain clipped and reconstruction error is 13.2 times the endpoint-residual baseline. The results identify robust default choices and, equally importantly, configurations that complete without exceptions but are numerically unusable.

Keywords: partial differential equations, flow matching, physics guidance, inverse problems, ablation study, numerical stability

1 Introduction

Physics-guided generative inference contains several interacting numerical choices that are not determined by the trained model alone. Guidance can be computed from observations, a PDE residual, or both; a loss can be evaluated at the current noisy state, the next state, or a denoised endpoint; and the integration path can be stochastic, deterministic, or hybrid. The number and placement of observations, the time grid, and the gradient clip introduce further problem-dependent behavior. A configuration may therefore finish successfully while producing a reconstruction many orders of magnitude worse than a neighboring configuration.

The purpose of this ablation study is to separate these effects under a common evaluation protocol. We ask five questions. First, which guidance component supplies the information needed for forward, inverse, and joint reconstruction? Second, how do the sampler phase and loss-evaluation state affect stability? Third, do more time steps or a higher-order integrator reliably improve accuracy? Fourth, how sensitive is each PDE to observation density, placement, and noise? Fifth, does tighter gradient clipping merely prevent exceptions, or does it also restore useful Navier–Stokes solutions?

The main empirical conclusion is that there is no single scalar “best” configuration. Observation guidance, endpoint loss evaluation, Euler integration, and a stochastic component are robust choices across the three problems. In contrast, the best hybrid schedule and observation pattern are problem

dependent, and lower reconstruction error can coincide with a larger PDE residual. The study is descriptive rather than causal: most individual configurations use one random seed, while five-seed runs are available only for the designated stability group.

2 Experimental Design and Data Processing

2.1 Ablation scope

The experiment varies ten common factors: guidance components, loss state, sampler phase, time grid, number of steps, integration method, observation count, observation layout, observation noise, and random seed. The Navier–Stokes suite additionally compares three temporal residual modes. Table 1 gives the resulting unique run counts. Forward runs score the solution field, inverse runs score the coefficient field, and joint runs score both fields. Burgers is represented by one output channel, so its two stored relative-error fields are identical.

The reference joint configuration uses observation and PDE guidance, an endpoint loss, stochastic sampling, a uniform 100-step grid, Euler updates, 500 per-sample random observations, and zero observation noise. Ablation groups modify one design dimension around this reference. Identical anchor configurations intentionally recur in different groups; they are retained because they provide within-group comparisons and are not historical duplicates.

2.2 Run identity and removal of historical duplicates

The historical aggregate contains 355 final-metric records. We define run identity by the composite key

$$(\text{PDE}, \text{task}, \text{ablation_name}).$$

Within each identity, we retain the most recent run satisfying `status=ok`, a non-error PDE evaluation status, and finite values for RelL2_a , RelL2_u , and L_{PDE} . This procedure produces 274 unique records and excludes 81 superseded records. All 81 exclusions are older Navier–Stokes joint runs with `clip_threshold=1010`; Poisson and Burgers have no historical rerun duplicates. The historical file remains unchanged for auditability, while the paper uses only `summary_latest_unique.csv`. The complete exclusions and their selected replacements are recorded in `summary_excluded_runs.csv`.

The selected records all pass status, step-count, sample-count, and finite metric checks. Resolved clipping thresholds are 50 for all Burgers runs; 10^{10} for Poisson joint/forward and Navier–Stokes forward; 50 for Poisson and Navier–Stokes inverse runs; and 10^4 for Navier–Stokes joint runs.

2.3 Metrics and comparison rules

Every reported configuration contains five primitive metrics: coefficient-field relative error RelL2_a , solution-field relative error RelL2_u , coefficient observation loss $L_{\text{obs},a}$, solution observation loss $L_{\text{obs},u}$, and PDE loss L_{PDE} . The relative errors are mean per-sample relative L_2 errors; the observation losses are the final unweighted masked MSE values against the actual observations, including configured observation noise. Clean-target observation losses remain available in the aggregate files as diagnostics.

Table 1: Scope of the deduplicated experiment. Counts refer to unique analysis-ready runs after historical reruns are removed.

Problem	Forward	Inverse	Joint	Total
Poisson	4	4	85	93
Burgers	0	0	85	85
Non-bounded Navier–Stokes	4	4	88	96
Total	8	8	258	274

For ranking joint Poisson and Navier–Stokes runs only, we define the balanced reconstruction index

$$E_{\text{bal}} = \frac{\text{RelL2}_a + \text{RelL2}_u}{2}. \quad (1)$$

For forward and inverse tasks, the target score is RelL2_u and RelL2_a , respectively. For Burgers, the target score is the single-channel relative error. Equal weighting in Equation 1 is an analysis convention rather than a new training objective, and the index never replaces the separately reported a - and u -field errors. The complete one-row-per-run reporting view is `ablation_report_metrics.csv`; its stable metric columns are `rel_l2_a`, `rel_l2_u`, `L_obs_a`, `L_obs_u`, and `L_pde`.

The reported L_{PDE} is used as a physical-consistency diagnostic only when the residual definition is unchanged. It must not be compared directly across endpoint secant, Hermite bridge, and near-endpoint temporal modes, which use different residual channels and normalizations. Unless a result is drawn from the five-seed stability group, values represent one run with batch size one and should be interpreted as effect-size diagnostics rather than formal statistical estimates.

3 Observation Guidance Supplies the Reconstruction Signal

Across every problem and task, observation guidance reduces the target error by approximately one to two orders of magnitude relative to no guidance. Table 2 reports the complete guidance-component sweep. PDE guidance alone is nearly indistinguishable from no guidance. Adding the PDE term to observation guidance produces only a small forward-task improvement for Navier–Stokes and no consistent improvement elsewhere.

The result does not imply that the PDE term is intrinsically useless. It shows that, under the tested weights and residual definitions, the observation loss already provides nearly all of the reconstruction signal. Moreover, a small PDE residual alone is not sufficient for identification: Burgers PDE-only guidance obtains $L_{\text{PDE}} = 6.43 \times 10^{-4}$, smaller than the observation-guided baseline, while its reconstruction error is 1.061 rather than 0.016. The PDE admits many low-residual fields that are not the target field; observations select among those fields.

Table 2: Complete primitive metrics for the guidance-component ablation. NS denotes non-bounded Navier–Stokes. Observation losses are reported for both fields even when the task gate activates only one of them.

Problem	Task	Guidance	RelL2 _a	RelL2 _u	$L_{\text{obs},a}$	$L_{\text{obs},u}$	L_{PDE}
Poisson	Forward	No guide	0.706328	0.636149	0.0475036	8.30383×10^{-6}	0.00501552
Poisson	Forward	PDE only	0.706371	0.636293	0.0475095	8.30763×10^{-6}	0.00501407
Poisson	Forward	Obs. only	0.104347	0.0567385	0.000759209	7.49590×10^{-8}	0.00493408
Poisson	Forward	Obs.+PDE	0.104354	0.0567902	0.000759369	7.50893×10^{-8}	0.00492117
Poisson	Inverse	No guide	0.706328	0.636149	0.0475036	8.30383×10^{-6}	0.00501552
Poisson	Inverse	PDE only	0.706474	0.636608	0.0475234	8.31597×10^{-6}	0.00499225
Poisson	Inverse	Obs. only	0.184290	0.0400967	0.00369552	3.34923×10^{-8}	0.00495474
Poisson	Inverse	Obs.+PDE	0.184284	0.0400938	0.00369519	3.34877×10^{-8}	0.00493661
Poisson	Joint	No guide	0.706328	0.636149	0.0475036	8.30383×10^{-6}	0.00501552
Poisson	Joint	PDE only	0.706365	0.636290	0.0475089	8.30756×10^{-6}	0.00501193
Poisson	Joint	Obs. only	0.0987497	0.0335385	0.000674767	2.47046×10^{-8}	0.00486790
Poisson	Joint	Obs.+PDE	0.0987495	0.0335390	0.000674772	2.47065×10^{-8}	0.00485743
Burgers	Joint	No guide	1.08051	1.08051	0.109369	0.106683	0.000746940
Burgers	Joint	PDE only	1.06139	1.06139	0.105655	0.103229	0.000643136
Burgers	Joint	Obs. only	0.0161125	0.0161125	2.20532×10^{-5}	2.04047×10^{-5}	0.00304960
Burgers	Joint	Obs.+PDE	0.0160872	0.0160872	2.19834×10^{-5}	2.03668×10^{-5}	0.00298843
NS	Forward	No guide	1.20948	0.997277	0.0549511	0.0441466	0.00243790
NS	Forward	PDE only	1.20948	0.997271	0.0549515	0.0441461	0.00243838
NS	Forward	Obs. only	0.214954	0.124017	0.00138949	0.000725228	0.00276624
NS	Forward	Obs.+PDE	0.213913	0.122684	0.00138016	0.000716983	0.00271786
NS	Inverse	No guide	1.20948	0.997277	0.0549511	0.0441466	0.00243790
NS	Inverse	PDE only	1.20952	0.997295	0.0549539	0.0441498	0.00243765
NS	Inverse	Obs. only	0.342449	0.206961	0.00473021	0.00162207	0.00273852
NS	Inverse	Obs.+PDE	0.346227	0.209956	0.00484030	0.00166886	0.00273813
NS	Joint	No guide	1.20948	0.997277	0.0549511	0.0441466	0.00243790
NS	Joint	PDE only	1.20943	0.997223	0.0549473	0.0441453	0.00243845
NS	Joint	Obs. only	0.199914	0.115232	0.00111870	0.000584215	0.00277407
NS	Joint	Obs.+PDE	0.199993	0.115158	0.00112182	0.000581448	0.00277131

4 Stochastic-to-Deterministic Sampling Improves Accuracy but Can Harm the Residual

Endpoint loss evaluation with a stochastic component is the only setting that is reliable across all three problems. Evaluating guidance at `xt` or `x_next` raises the stochastic target score from 0.066 to 0.346/0.257 on Poisson, from 0.016 to 0.629/0.481 on Burgers, and from 0.158 to 0.359/1.153 on Navier–Stokes. Replacing stochastic sampling with a deterministic endpoint update produces scores of 2.56×10^4 , 2.63×10^3 , and 1.36×10^5 , respectively.

Hybrid schedules can outperform the stochastic reference, but their optimal switch points differ by problem. Table 3 and Figure 1 compare the best sampler-phase and time-grid variants with the reference joint score. Burgers benefits most: `hybrid_s2d` with switch ratio 0.2 reduces the error by 66.3%. Poisson also improves, but its best phase setting multiplies L_{PDE} by 166. Navier–Stokes obtains its best reconstruction from `cosine + hybrid_s2d`, with a 32.0% score reduction and a 13.9-fold residual increase.

Table 3: Primitive metrics for the stochastic reference and the settings selected by the sampler-phase and time-grid ranking indices.

Problem	Source	Setting	RelL2 _a	RelL2 _u	L _{obs,a}	L _{obs,u}	L _{PDE}
Poisson	Reference	Stochastic	0.0987495	0.0335390	0.000674772	2.47065×10^{-8}	0.00485743
Poisson	Phase	hybrid_s2d, 0.2	0.0858454	0.0151030	0.000580406	4.34068×10^{-9}	0.804162
Poisson	Grid	Cosine + hybrid_s2d	0.0895861	0.0194788	0.000569167	8.42302×10^{-9}	0.0271510
Burgers	Reference	Stochastic	0.0160872	0.0160872	2.19834×10^{-5}	2.03668×10^{-5}	0.00298843
Burgers	Phase	hybrid_s2d, 0.2	0.00541692	0.00541692	2.81017×10^{-6}	2.75459×10^{-6}	0.00667524
Burgers	Grid	Cosine + hybrid_s2d	0.00552473	0.00552473	3.12696×10^{-6}	2.08008×10^{-6}	0.00273157
Navier–Stokes	Reference	Stochastic	0.199993	0.115158	0.00112182	0.000581448	0.00277131
Navier–Stokes	Phase	hybrid_s2d, 0.5	0.183344	0.0850448	0.000875568	0.000297114	0.00669611
Navier–Stokes	Grid	Cosine + hybrid_s2d	0.156057	0.0581717	0.000575255	0.000126354	0.0385353

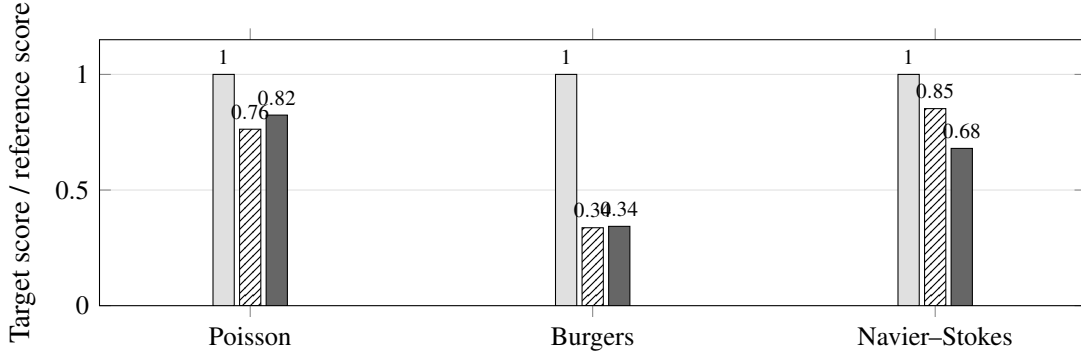


Figure 1: Best reconstruction scores normalized by the joint stochastic reference for each PDE. Values below one indicate an accuracy improvement. Within each PDE, the bars from left to right show the reference, best phase sweep, and best grid sweep. The figure is a ranking visualization only; Table 3 reports both field errors and all loss components separately.

The common mechanism is consistent with early stochastic exploration followed by late deterministic refinement. Beginning deterministically and switching to stochastic sampling is much less reliable: hybrid_d2s with switch ratio 0.8 produces errors of 3334 on Poisson, 343 on Burgers, and 17744 on Navier–Stokes. The exact switch point is nevertheless problem specific, which argues against treating a hybrid schedule as a universal default.

5 More Steps and Midpoint Integration Do Not Give Universal Gains

Step-count effects differ across the PDEs. In the step sweep, Poisson reaches its lowest score at 2000 stochastic steps (0.05953), compared with 0.06614 at 100 steps, but wall-clock time grows from approximately 9 to 179 seconds. Burgers reaches its stochastic minimum at 1000 steps (0.00815) and slightly regresses at 2000 steps (0.00868); the 50-step hybrid_s2d result is already 0.00992. Navier–Stokes is qualitatively different: hybrid_s2d is best at 100 steps (0.13453), and additional steps do not improve it; stochastic sampling also worsens beyond 100 steps. Consequently, the

number of steps is a problem- and schedule-dependent regularization choice rather than a monotone accuracy control.

Midpoint integration increases 100-step runtime by factors of 1.50 on Poisson, 1.50 on Burgers, and 1.66 on Navier–Stokes. Its reconstruction changes are small and inconsistent. For example, midpoint slightly improves Navier–Stokes `hybrid_s2d` from 0.13336 to 0.13279, but raises L_{PDE} from 0.00689 to 0.01066. The change is smaller than the five-seed variability and does not justify the additional compute as a default. Euler is therefore the more defensible baseline.

6 Observation Requirements Are Strongly Problem Dependent

Observation density is the largest source of cross-problem heterogeneity. Figure 2 normalizes each sparsity result by its 500-observation score. Poisson shows a sharp transition: 250 observations are 12.9 times worse than 500, and 50 observations are 100 times worse. Navier–Stokes is also unstable at 50–100 observations but is within 17% of the 500-observation result at 250. Burgers degrades smoothly; 50 observations approximately double the error rather than causing an order-of-magnitude failure. Increasing from 500 to 1000 observations produces no material gain for any problem.

Placement is similarly problem dependent. Random and per-sample random masks are best for Poisson (score 0.06614), whereas a fixed mask is poor (1.150). Burgers obtains a promising fixed-mask score of 0.00592, compared with 0.01609 for per-sample random masks, but this is a single-mask result and may reflect a favorable placement. Navier–Stokes prefers a grid (0.15542); fixed and sensor-column layouts degrade to 0.854 and 0.477. These results support problem-specific mask design rather than a single universal sensor policy.

Noise sensitivity also differs. At 10% noise, the target score changes by +4.4% for Poisson, +63.9% for Burgers, and −0.2% for Navier–Stokes relative to the corresponding zero-noise run. The small apparent Navier–Stokes improvement is below seed variation and should be interpreted as no detectable degradation, not as a beneficial effect of noise.

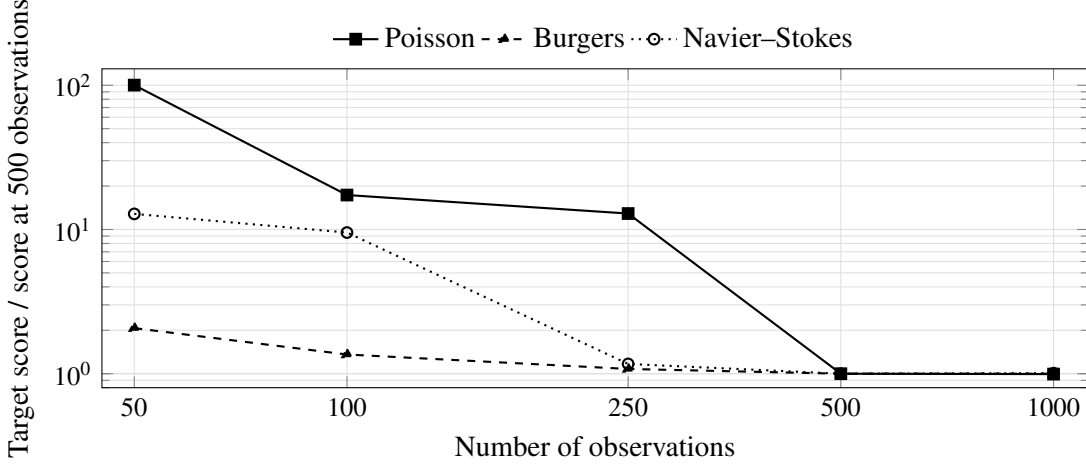


Figure 2: Sensitivity to observation sparsity. Ratios are computed within each PDE relative to its 500-observation run; both axes are logarithmic. Line style and marker shape duplicate the series distinction for grayscale readability.

7 Stability Checks Separate Runtime Success from Numerical Success

Five-seed results quantify baseline variability (Table 4). Navier–Stokes is the most repeatable in reconstruction score, with a 2.05% coefficient of variation. Burgers has 15.1% relative variability. Poisson’s balanced score varies by 12.5%, while its solution error and PDE residual have coefficients of variation of 37.1% and 49.0%, respectively. Small single-run differences—including most observation-only versus observation+PDE differences—are therefore not statistically distinguishable under the available replication.

Every one of the 274 selected records has finite final metrics, but 57 configurations have a target score of at least 10: 20 Poisson, 17 Burgers, and 20 Navier–Stokes runs. These cases are concentrated in deterministic sampling, late `hybrid.d2s` switches, and incompatible time grids. We classify them as numerical divergence despite `status=ok`. This distinction is important: exception-free execution is a software property, whereas a useful reconstruction is a numerical property.

The Navier–Stokes Hermite residual illustrates a third failure mode. With a clip threshold of 10^{10} , the original run records five steps and then terminates because the PDE loss contains NaN/Inf. Repeating the run at 10^4 completes all 100 steps, but clipping is active for 88 steps and the minimum gradient scale is 0.054. Its balanced score is 2.078, compared with 0.158 for endpoint secant, a 13.2-fold degradation. Tighter clipping thus solves the runtime exception but not the underlying stiffness or objective mis-scaling.

Table 4: Five-seed stability of the reference configuration. Each cell is mean $\pm$ sample standard deviation; no balanced score replaces the two field-level errors.

Problem	RelL2 _a	RelL2 _u	L _{obs,a}	L _{obs,u}	L _{PDE}
Poisson	0.10555 $\pm$ 0.00598	0.03890 $\pm$ 0.01442	7.80 $\times 10^{-4} \pm 1.10 \times 10^{-4}$	3.45 $\times 10^{-8} \pm 2.72 \times 10^{-8}$	0.003744 $\pm$ 0.001836
Burgers	0.01458 $\pm$ 0.00220	0.01458 $\pm$ 0.00220	1.89 $\times 10^{-5} \pm 5.63 \times 10^{-6}$	1.51 $\times 10^{-5} \pm 3.77 \times 10^{-6}$	0.003354 $\pm$ 0.000523
Navier–Stokes	0.19816 $\pm$ 0.00415	0.11696 $\pm$ 0.00357	0.001082 $\pm$ 0.000096	0.000577 $\pm$ 0.000014	0.003056 $\pm$ 0.000199

8 Discussion and Practical Recommendations

The three PDE suites support four general conclusions.

Observations identify the solution; the tested PDE term mainly regularizes it. PDE-only guidance does not recover the target even when its residual is small. This behavior is expected when many fields approximately satisfy the operator but only a subset agrees with the observed coefficient or solution. The present weights make the PDE term a weak discriminator after observations are available. A stronger claim would require re-tuning the PDE coefficient and replicating the comparison across seeds.

Stochasticity is a stability mechanism. Pure deterministic sampling is consistently catastrophic. A late deterministic phase can improve accuracy, but only after stochastic sampling has kept the trajectory in a stable region. We recommend endpoint loss, Euler integration, and a stochastic reference as the default. A problem-specific `hybrid.s2d` schedule should be treated as a tuned alternative: ratio 0.2 is promising for Poisson and Burgers, while ratio 0.5 with a cosine grid is the strongest Navier–Stokes accuracy candidate.

Accuracy and physical residual form a Pareto problem. The best reconstruction setting is not always the setting with the smallest L_{PDE} . Poisson and Navier–Stokes gain accuracy at a substantial residual cost, while Burgers cosine + `hybrid.s2d` improves both quantities relative to the reference. Future model selection should report a two-objective Pareto front rather than rank configurations by reconstruction error alone.

Observation policy must follow the operator. For the tested resolution and guidance weights, 500 observations are a safe cross-problem default. Poisson should not be reduced below this value without retuning; Navier–Stokes may tolerate 250 but is unstable at 100; Burgers is more tolerant but more sensitive to noise. Fixed or structured masks require multiple-mask replication before they can be recommended.

On current evidence, we advise against pure deterministic sampling, large `hybrid.d2s` switch ratios, and the present Hermite residual. Midpoint integration does not offer a favorable accuracy–cost tradeoff. The strongest candidates for a confirmatory study are Burgers `hybrid.s2d` at ratio 0.2 and Navier–Stokes cosine + `hybrid.s2d` at ratio 0.5, each evaluated with at least five seeds and with reconstruction and physical residual reported jointly.

9 Limitations

This study has five main limitations. First, most configurations contain one sample and one seed; only the stability group has five independent seeds. Second, the study is one-factor-at-a-time rather

than fully factorial, so the best settings from different groups cannot be assumed to combine additively. Third, selecting the best run among many single-seed alternatives introduces selection bias and multiple-comparison risk. Fourth, the equal-weight balanced score may not reflect application-specific costs for coefficient and solution errors. Fifth, L_{PDE} is residual-definition dependent and does not measure all forms of physical plausibility.

There is also a reproducibility hazard in the current repository state. The selected Navier–Stokes joint runs resolve to a clip threshold of 10^4 , but `configs/main/both/nsnonbounded.yaml` still contains 10^{10} . The stable threshold should be committed to a canonical configuration or passed through an explicit, recorded override before the sweep is rerun.

10 Conclusion

The deduplicated 274-run study shows that observation guidance, endpoint loss evaluation, and stochastic sampling are the most transferable design choices across Poisson, Burgers, and non-bounded Navier–Stokes problems. Hybrid stochastic-to-deterministic sampling can improve reconstruction substantially, but the optimal switch is problem dependent and can increase the PDE residual. More steps and higher-order integration are not uniformly beneficial, and observation density has distinct thresholds across operators. Finally, gradient clipping should be interpreted as a safety mechanism rather than a guarantee of useful solutions: it prevents the Hermite residual from crashing but leaves the method heavily clipped and inaccurate. These findings narrow the stable design space and define the multi-seed, multi-objective experiments needed for a defensible final configuration.

Appendix A. Complete Group-Level Summary

The complete group-level report is emitted as `ablation_report_metrics.csv`. It contains one selected row for every (PDE,task,ablation_name) identity, including all 274 currently completed configurations. Unlike the previous appendix, it does not collapse RelL2_a and RelL2_u into a target score and does not omit the observation losses. Table 5 defines its stable metric contract; configuration columns and source paths accompany these values in the CSV.

For repeated configurations, `summary_latest_run_seed_grouped.csv` reports mean, sample standard deviation, standard error, 95% interval half-width, median, p90, minimum, and maximum separately for all five fields.

Table 5: Required columns in the complete ablation reporting view.

CSV column	Symbol	Definition
rel_l2_a	RelL2_a	Coefficient-field relative L_2 error
rel_l2_u	RelL2_u	Solution-field relative L_2 error
L_obs_a	$L_{\text{obs},a}$	Final masked observation MSE for a
L_obs_u	$L_{\text{obs},u}$	Final masked observation MSE for u
L_pde	L_{PDE}	Final evaluation PDE loss

Appendix B. Data Quality and Reproducibility Artifacts

The analysis is backed by the following files in `outputs/ablations`:

- `summary_all_raw.csv`: all 355 historical final records;
- `summary_latest_unique.csv`: the 274-record analysis snapshot;
- Complete reporting view with separate RelL2_a , RelL2_u , $L_{\text{obs},a}$, $L_{\text{obs},u}$, and L_{PDE} : `ablation_report_metrics.csv`;
- `summary_excluded_runs.csv`: the 81 superseded records and their replacements;
- `summary_latest_run_seed_grouped.csv`: grouped run-level statistics for the same five primitive metrics;
- `metrics_per_sample_latest_unique.csv`: per-sample current records; and
- `curves_latest_grouped.csv`: step-level curves restricted to selected runs.

The supporting code is also auditable:

- `sampling/aggregate.py` implements the latest-snapshot selector; and
- `tests/test_aggregate_results.py` verifies that historical rows are preserved while superseded rows are excluded from the current snapshot.

The selector uses a deterministic recency tie-break and emits an explicit exclusion reason for every discarded final record.